



Machine Learning: Fungi Image Classification

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Introduction

This project develops a machine learning model to classify microscopic fungi images from the DeFungi dataset into five categories (H1, H2, H3, H5, H6). Using image preprocessing, feature extraction, and dimensionality reduction with PCA, the study compares multiple classifiers to identify the most effective approach. The results demonstrate that supervised learning can support accurate and consistent fungi classification from image data.

Background

Classification

Fungi classification from microscopic images is an important task in biological and agricultural analysis, as it supports the detection of infections and improves diagnostic efficiency. Traditionally, identification relies on manual inspection, which can be time-consuming and subject to variability. Machine learning provides an alternative by learning patterns directly from image data, enabling faster and more consistent classification of fungal categories.

Methodology

Preprocessing

The process began by organizing the DeFungi images into a structured dataset with file paths and labels. Images were filtered for valid formats, resized to 32×32 pixels, converted to RGB, normalized, and flattened into numerical feature vectors. Class labels were encoded into numeric values, and data augmentation was applied to balance the dataset across all five classes.

Model Training

Features were standardized and reduced using Principal Component Analysis (PCA) to retain 95% of the variance while lowering dimensionality. The dataset was then split into training, validation, and test sets using stratified sampling. Multiple models were trained and compared, and hyperparameter tuning with cross-validation was used to optimize the strongest candidates.

Model	Training CV F1 Weighted	Test Accuracy	Test Precision Weighted	Test Recall Weighted	Test F1 Weighted
LGBMClassifier	0.7339	0.7598	0.7556	0.7598	0.753
XGBClassifier	0.7113	0.7398	0.7387	0.7398	0.7288
RandomForestClassifier	0.6448	0.6798	0.6716	0.6798	0.6599
BaggingClassifier	0.6177	0.6553	0.6499	0.6553	0.6284
DecisionTreeClassifier	0.5777	0.6122	0.6135	0.6122	0.6036
NearestCentroid	0.4638	0.4737	0.4699	0.4737	0.4587

Table 1. Tuned Model Performance

Results

Performance Comparison

The results show that LGBMClassifier achieved the best overall performance, with the highest weighted F1-score and test accuracy among all evaluated models. XGBClassifier followed closely, while Random Forest and other models performed moderately, and simpler methods such as Decision Tree and Nearest Centroid showed lower performance.

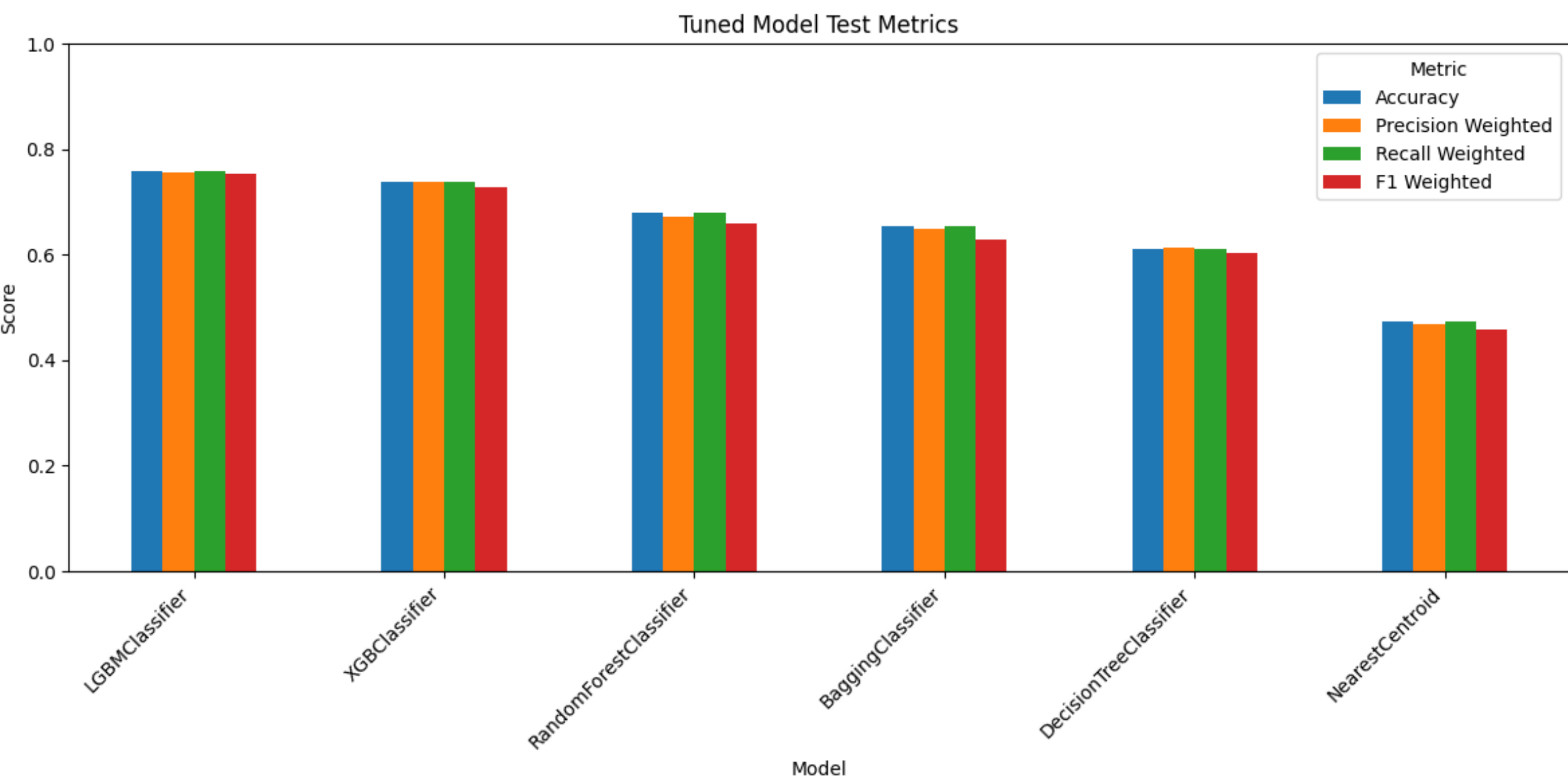


Figure 1. Tuned model Test Metrics

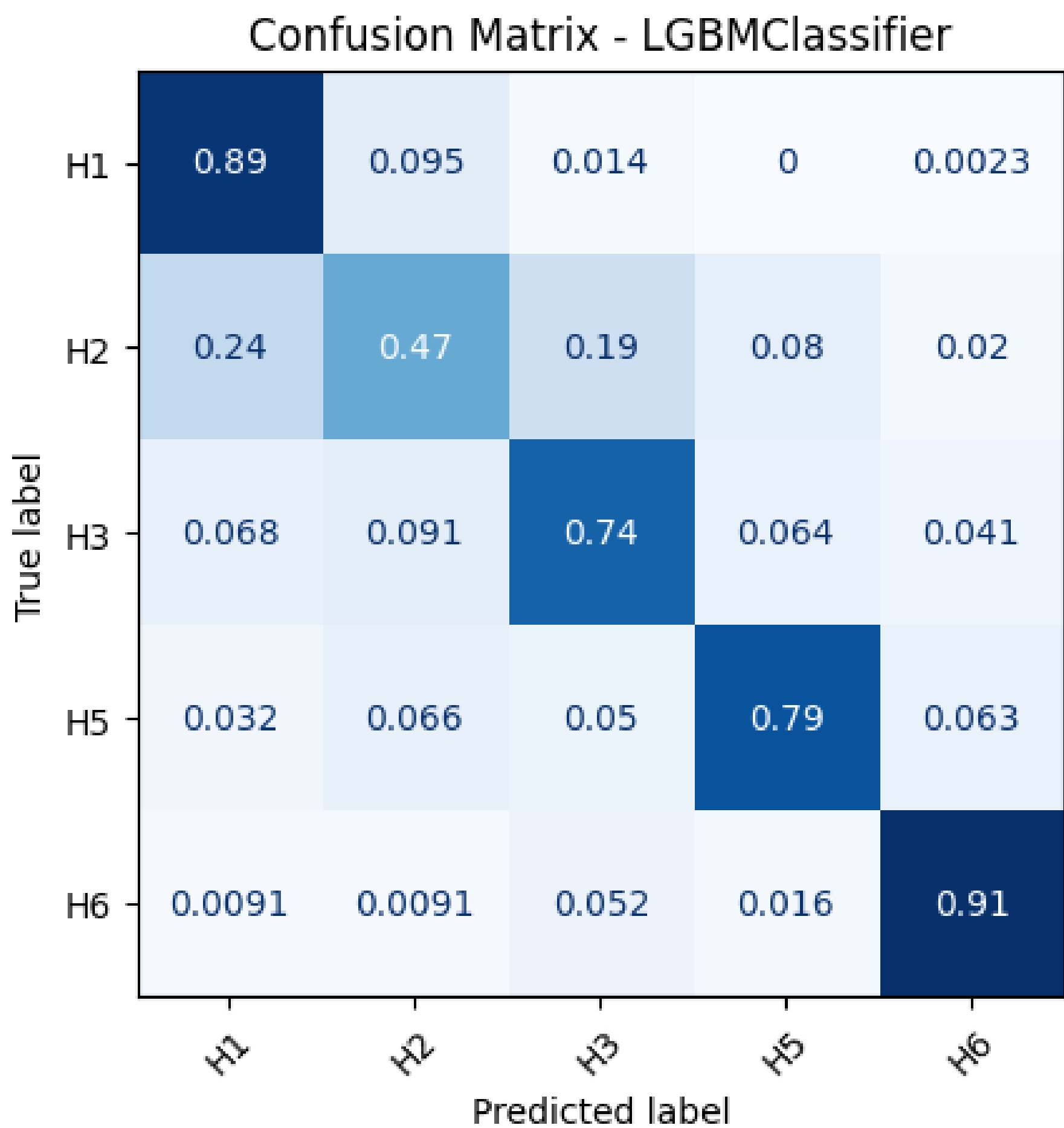


Figure 2. Normalized Confusion Matrix

Key Findings

Overall, the results indicate that boosting-based models are most effective for this classification task. While performance was strong for some classes, others were more difficult to distinguish, highlighting challenges related to class similarity and dataset complexity.

Conclusions

This project demonstrates that machine learning can effectively classify fungi images using the DeFungi dataset. Boosting-based models, particularly LightGBM, achieved the best performance, showing strong predictive capability. While results are promising, challenges such as class overlap remain, highlighting opportunities for improvement with more advanced techniques and refined preprocessing.

References

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